FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Diarrhoea, Vomiting & Dehydration

Tutorial | Week 2 · Diarrhoea, fluids, HIV & anaemia | Paediatrics | Dr Nabbale

Diarrhoea (3 or more loose or watery stools in 24 hours) kills more Ugandan under-fives than almost anything except pneumonia and malaria; you save these children by grading dehydration correctly at the bedside and giving the right WHO plan (A, B or C) with the right volume, plus zinc, continued feeding and antibiotics only for blood in the stool or cholera.

Must know before the ward round

1. Diarrhoea = 3 or more loose or watery stools in 24 h. Acute < 14 days; persistent ≥ 14 days; dysentery = visible blood in the stool.
2. Dehydration signs (any 2): severe = lethargic/unconscious, sunken eyes, unable to drink or drinks poorly, skin pinch goes back very slowly (> 2 s); some = restless/irritable, sunken eyes, drinks eagerly/thirsty, skin pinch goes back slowly.
3. Plan A (no dehydration): 4 rules = extra fluid (ORS 50–100 mL after each loose stool if < 2 y, 100–200 mL if 2 y or older), zinc, continue feeding, when to return. Give 2 ORS packets.
4. Plan B (some dehydration): ORS 75 mL/kg over 4 hours in the clinic, then reassess and reclassify.
5. Plan C (severe): Ringer's lactate 100 mL/kg. Under 12 months: 30 mL/kg in 1 h then 70 mL/kg in 5 h. 12 months and older: 30 mL/kg in 30 min then 70 mL/kg in 2.5 h. Repeat the first 30 mL/kg once if the radial pulse is still very weak.
6. No IV possible: ORS by nasogastric tube 20 mL/kg/h for 6 h (120 mL/kg); refer if not improving in 3 h.
7. Zinc (UCG 2023): under 6 months 10 mg, 6 months or older 20 mg once daily for 10 days. WHO 2024 now suggests 5 mg daily for 10–14 days for all children up to 10 years (fewer vomits); follow the national dose on the ward.
8. Low-osmolarity ORS: Na 75, glucose 75, K 20, Cl 65, citrate 10 mmol/L; total osmolarity 245 mOsm/L.
9. No antibiotics for watery diarrhoea (WHO 2024). Dysentery: ciprofloxacin 15 mg/kg orally 12-hourly for 3 days; ceftriaxone 50–80 mg/kg IV/IM daily for 3 days if severely ill.
10. Cholera in children (Uganda MoH 2025): doxycycline 2–4 mg/kg orally single dose; alternative azithromycin 20 mg/kg single dose (max 1 g).
11. SAM changes everything: no IV fluids unless shocked; ReSoMal 5 mL/kg every 30 min for 2 h, then 5–10 mL/kg/h for up to 10 h; shock = 15 mL/kg over 1 h of glucose-containing fluid.
12. Never give antidiarrhoeals (loperamide) or antiemetics (chlorpromazine) to young children with diarrhoea; never give plain 5% dextrose to rehydrate.

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In a nutshell

Diarrhoea is the passage of **3 or more loose or watery stools in 24 hours**. In children it is usually an infection of the gut (rotavirus, *E. coli*, *Shigella*, *Cryptosporidium*, cholera), and it kills by **dehydration**, by **dysentery and sepsis**, and by **worsening malnutrition**. Vomiting on its own usually means something else (malaria, UTI, meningitis, a surgical abdomen), but vomiting with diarrhoea is just gastroenteritis until proven otherwise.

The bedside skill is simple and life-saving: **assess the degree of dehydration with four signs, choose Plan A, B or C, give the exact volume, and reassess**. Add **zinc**, **keep feeding**, and give antibiotics **only** for blood in the stool, cholera or a separate bacterial infection.

> **CLINICAL PEARL**
> The one thing never to forget: **first check whether the child is severely malnourished.** In a wasted or oedematous child, Plan C can kill: severe dehydration is over-diagnosed in SAM, the heart cannot handle a fast litre, and you must use ReSoMal slowly instead. Measure MUAC and look at both feet **before** you hang the drip.

Definitions & classification

Term	Definition
Diarrhoea	3 or more loose or watery stools in 24 hours (a change from normal; frequent pasty stools in a breastfed baby are **not** diarrhoea)
Acute watery diarrhoea	Diarrhoea lasting **less than 14 days** without visible blood (includes cholera)
Dysentery	Diarrhoea with **visible blood** in the stool (usually *Shigella*)
Persistent diarrhoea	An episode that began acutely and lasts **14 days or more**
Severe persistent diarrhoea	Persistent diarrhoea **with dehydration**: hospital management
Dehydration	Loss of water and salts; graded as **no**, **some** or **severe** (IMCI/WHO), roughly <5%, 5–10% and >10% of body weight
Hypovolaemic shock	All three of: **cold extremities, capillary refill > 3 s, and a weak, fast pulse** (WHO ETAT 2016); with severe dehydration it needs Plan C immediately
Isonatraemic / hyponatraemic / hypernatraemic dehydration	Serum Na 130–150 / < 130 / > 150 mmol/L

IMCI and WHO classification of dehydration (2 or more signs)

Sign	No dehydration	Some dehydration	Severe dehydration
General condition	Well, alert	**Restless, irritable**	**Lethargic or unconscious**
Eyes	Normal	Sunken	Sunken
Fontanelle (infant)	Normal	Sunken	Very sunken
Drinking	Drinks normally	**Thirsty, drinks eagerly**	**Drinks poorly or not able to drink**
Skin pinch (abdomen)	Goes back immediately	Goes back **slowly** (< 2 s)	Goes back **very slowly** (> 2 s)
Plan	**A**	**B**	**C**

Bold signs are the "key" signs. "Sunken eyes" counts in both; always ask the mother whether the eyes look different from usual.

Epidemiology & Uganda context

- **Globally**, diarrhoea is the **third leading cause of death in children under 5**: about **444,000 under-five deaths a year**, plus about 51,000 children aged 5–9 (WHO fact sheet). Pneumonia and diarrhoea together cause **23% of under-five deaths** (WHO 2024 guideline).
- **Uganda**: diarrhoea is one of the "big three" with pneumonia and malaria in every district report. Peaks follow the rains and **cholera outbreaks** recur in border districts (Kasese, Busia, Mbale, Namayingo), refugee settlements and Kampala's informal settlements.
- **Rotavirus vaccine** (two doses at **6 and 10 weeks** in the UNEPI schedule) has cut severe rotavirus admissions; measles–rubella vaccine and vitamin A also prevent diarrhoeal deaths.
- **Malnutrition and diarrhoea feed each other**: diarrhoea causes weight loss and zinc loss; malnourished children have more, longer and deadlier episodes. UDHS 2022: **26% of under-fives stunted, 2.9% wasted**.

> **UGANDA CONTEXT**
> At **HC II** you can give Plans A and B (ORS and zinc are HC II items in UCG 2023). **HC III** can start Plan C (IV Ringer's lactate). Many HC IIIs and IVs lack giving sets, cannulas for infants or Ringer's lactate at night, so the **nasogastric ORS route (20 mL/kg/h for 6 h)** is your back-up, and the referral note must say exactly how much fluid was given and when. Mulago, Mengo and the regional referral hospitals have ORS corners ("ORT corners") where Plan B is supervised; use them.

Pathophysiology

How the gut loses water

- The small bowel secretes and absorbs about **litres** of fluid a day. Most of it is reabsorbed by sodium-coupled transport.
- **Secretory diarrhoea** (cholera toxin, enterotoxigenic *E. coli*): toxin raises cAMP in crypt cells → chloride and water pour out. Stools are huge and watery ("rice water"). The villi are intact.
- **Osmotic / malabsorptive diarrhoea** (rotavirus damaging villous tips, lactose intolerance, F-75 in some SAM children): unabsorbed solute holds water in the lumen.
- **Inflammatory (invasive) diarrhoea** (*Shigella*, *Campylobacter*, *Salmonella*, *E. histolytica*): mucosal invasion → blood, mucus, fever, tenesmus, and risk of sepsis.

Why ORS works

Sodium–glucose co-transport (SGLT1) in the villous enterocytes keeps working even when cholera toxin is driving secretion. One glucose molecule carries sodium in with it, and **water follows the sodium**. That is why the ratio matters: **low-osmolarity ORS** (glucose 75 and Na 75 mmol/L, total **245 mOsm/L**) is absorbed better than the old 311 mOsm/L formula, cuts stool output and vomiting, and reduces the need for unscheduled IV fluids. Too much sugar (sweet drinks, sodas, glucose water) does the opposite: it pulls water **into** the gut.

Why children dehydrate so fast

- Higher **body water** (about 70–80% of weight in infants) and a much higher **turnover** per kg.
- Larger **surface area** for insensible loss; fever adds about 10% more fluid need per °C.
- They **cannot ask for a drink**, and they vomit what they are given.

Why zinc helps

Zinc deficiency is common in Ugandan children and **diarrhoea increases zinc loss**. Zinc restores mucosal repair, improves water and electrolyte absorption and immune function. It **shortens the episode and reduces stool volume**, and a full course reduces diarrhoea in the following months.

Sodium and potassium

- **Hyponatraemia** follows replacing salty losses with water or very dilute fluids (or hypotonic IV fluids). It causes lethargy and convulsions.
- **Hypernatraemia** follows high-solute feeds (over-concentrated formula), high fevers and water loss more than salt. Signs of dehydration are **masked** (the skin feels **doughy**), the child is **very irritable** and thirsty, and convulsions occur if it is corrected too fast.
- **Potassium** is lost in every stool. Hypokalaemia causes **floppiness, abdominal distension (ileus)** and arrhythmias. ORS contains 20 mmol/L of potassium.

Why SAM is different

In **severe acute malnutrition** the heart muscle is thin, sodium is retained in cells and potassium is low (failure of the sodium pump), and the kidneys cannot excrete a sodium load. Rapid IV fluid or high-sodium ORS leads to **heart failure and death**. The "dehydration" signs (sunken eyes, slow skin pinch) are present anyway in marasmus. Hence ReSoMal (Na 45, K 40 mmol/L) and slow oral rehydration.

Clinical acumen

Focused history

Ask	Why it matters
How long has the diarrhoea lasted?	≥ 14 days = persistent diarrhoea: think malnutrition, HIV, giardia, lactose intolerance
How many stools a day, how watery, how much?	Volume of loss; "rice water" = cholera
Blood or mucus in the stool?	Dysentery → antibiotic; red-currant jelly in an infant with pain = **intussusception**
Vomiting: how often, can the child keep anything down? **Bilious (green)?**	Persistent vomiting may need NG ORS; bilious vomiting = **obstruction** (surgical)
Drinking and urine: thirsty? last wet nappy?	Thirst = some dehydration; refusing to drink is a severe sign; anuria for > 6–8 h is late
Fever, convulsions, lethargy	Danger signs; malaria, meningitis, sepsis, hypoglycaemia, abnormal sodium
What has been given at home: ORS, herbs, tea, sodas, antibiotics, "*enema*"	Herbal enemas and purgatives cause severe illness; sugary drinks worsen diarrhoea
Feeding: breastfeeding? other milks? how diluted or concentrated?	Over-concentrated formula → hypernatraemia; dilute porridge → malnutrition
Contacts: others ill at home or in the village, cholera outbreak?	Cholera, *Shigella* outbreak, food poisoning
Water and sanitation: source of water, latrine, hand-washing	Prevention counselling and outbreak recognition
HIV status of child and mother, EID results, ART, co-trimoxazole	HIV causes persistent diarrhoea, cryptosporidium, and changes feeding advice

Ask	Why it matters
Immunisation: rotavirus (6 and 10 weeks), measles–rubella	Missed rota or measles raises risk; catch up before discharge
Recent measles	Measles-associated diarrhoea is severe; give vitamin A
Weight before the illness (child health card)	Weight loss gives the **real percentage dehydration**

Examination

1. **Emergency signs (ETAT ABCD)** first: obstructed breathing, severe respiratory distress, central cyanosis, **shock**, **coma**, **convulsing**, **severe dehydration** with lethargy.
2. **General**: alert, restless/irritable, or lethargic/unconscious? Is the child **floppy**? Sunken eyes (ask the mother)? Sunken fontanelle?
3. **Offer a drink** of ORS or water from a cup: drinks eagerly, poorly or not at all?
4. **Skin pinch**: pinch the abdominal skin halfway between the umbilicus and the side, **longitudinally**, hold 1 s and release. Very slow = > 2 s.
5. **Circulation**: hands warm or cold, **capillary refill** (press the thumbnail or sternum 3 s), radial pulse volume and rate.
6. **Breathing**: deep, sighing (Kussmaul) breathing = **metabolic acidosis** from bicarbonate loss; fast breathing with chest indrawing = pneumonia.
7. **Nutrition**: **weight**, length, **MUAC**, **oedema of both feet**, visible wasting, plot weight-for-length Z-score.
8. **Abdomen**: distension (ileus from hypokalaemia, obstruction, NEC), masses (intussusception "sausage"), tenderness, bowel sounds; **rectal prolapse** in *Shigella*.
9. **Others**: temperature, pallor, ears and throat, signs of HIV (oral thrush, lymphadenopathy, parotid enlargement), neck stiffness.

Red flags

> **RED FLAG**
> Act now (and call the senior) if: **lethargy or unconsciousness, unable to drink, convulsions, shock (cold hands + CRT > 3 s + weak fast pulse)**, bilious vomiting or a tender distended abdomen, **blood in the stool in an infant under 2 months**, severe acute malnutrition, **hypoglycaemia**, no urine for > 8 hours, a very irritable child with doughy skin (hypernatraemia), or **cholera** in an outbreak area (notify).

Thinking like a clinician

- **Grade dehydration with signs, not with the number of stools.** A child with 10 stools who is alert and drinking normally is Plan A.
- **"Some" dehydration is the commonest under-treated group.** Plan B means ORS **measured out in front of you** (75 mL/kg over 4 h), not two packets and home.
- **Vomiting is not a reason to give up on ORS.** Wait 10 minutes and give **small sips** (5 mL every 1–2 min by spoon or syringe). Most children keep ORS down once rehydration starts. If not, use an **NG tube**.
- **Deep breathing in a dehydrated child is acidosis**, not pneumonia. It settles with fluids.
- **A lethargic child after rehydration** is not "just tired": check **glucose**, then sodium, then think meningitis, malaria and sepsis.
- **Fever and diarrhoea in Uganda: test for malaria.** Malaria often presents with vomiting and loose stools in young children.
- **Blood plus pain plus pallor in an infant** = intussusception; blood plus oliguria plus pallor a week later = **haemolytic uraemic syndrome** (HUS).

- **If it lasts 14 days, it is no longer simple gastroenteritis.** Think HIV, malnutrition, giardia, TB, lactose intolerance.
- **Common student mistakes**: Plan C in a child with SAM; forgetting zinc; stopping breastfeeding; giving metronidazole "for diarrhoea"; not weighing the child, so the volume is guessed; not reassessing after the first 30 mL/kg.

Clinical workup

Step 0: First 5 minutes

1. **Triage (ETAT)**: emergency signs? Lethargic/unconscious, unable to drink, **shock**, convulsing. Severe dehydration is an **emergency sign** when the child is also lethargic.
2. **Weigh the child** (or use a weight estimate only if a scale is not available) and **measure MUAC and check both feet for oedema**. This decides the fluid plan.
3. **Airway and breathing**: recovery position if unconscious; oxygen if SpO_2 < 90%.
4. **Circulation**: cold hands, CRT > 3 s, weak fast pulse? **IV or IO access** (an intraosseous needle if two IV attempts fail).
5. **Glucose**: check RBS; if **< 3 mmol/L** (or no glucometer and the child is lethargic), give **10% glucose 5 mL/kg IV** (or 50 mL of sugar water orally/NG).
6. **Start fluids at once**: Plan C (no SAM) or the SAM shock protocol. If the child can drink, give ORS by cup while the line goes in.
7. **Call for help** if shocked, unconscious, convulsing or severely malnourished.

Step 1: Focused history

- [ ] **Duration** (persistent ≥ 14 days?) → different work-up and feeding plan.
- [ ] **Blood in stool?** → dysentery antibiotics; in an infant with colicky pain, think intussusception.
- [ ] **Vomiting frequency, bilious?** → NG ORS vs surgical review.
- [ ] **Urine output and last wet nappy** → severity; AKI risk.
- [ ] **Fever, convulsions** → malaria test, glucose, sodium, LP if meningism.
- [ ] **Fluids and remedies given at home** → herbal enemas, sugary drinks, inappropriate antibiotics.
- [ ] **Feeding history** (breastfeeding, formula concentration) → hypernatraemia; counselling.
- [ ] **HIV status** of mother and child; ART and co-trimoxazole → persistent diarrhoea causes; feeding advice.
- [ ] **Immunisation card**: rotavirus, measles–rubella → catch-up.
- [ ] **Contacts and water source** → cholera or *Shigella* outbreak; notify.
- [ ] **Pre-illness weight** from the child health card → real % deficit.

Step 2: Focused examination

General and vital signs (normal values by age):

Age	Respiratory rate (fast if ≥)	Heart rate (normal awake)
< 2 months	**60**/min	110–160
2–12 months	**50**/min	100–160
1–5 years	**40**/min	90–140
> 5 years	30/min	70–120

Then: **dehydration signs** (the four above plus fontanelle), **circulation** (CRT, pulses, temperature of hands), **acidotic breathing**, **weight and MUAC, oedema**, **abdomen** (distension, mass, tenderness, bowel sounds), **perianal area** (prolapse, excoriation), **signs of other infections** (ears, chest, neck stiffness), **signs of HIV**.

Document these positives	And these negatives
Weight, MUAC, WLZ, oedema grade	"No oedema of the feet; MUAC 13.2 cm"
Each dehydration sign separately	"No blood in stool seen"
CRT, pulse volume, hand temperature	"Abdomen soft, no mass, bowel sounds present"
Stool character seen yourself	"No neck stiffness; fontanelle not bulging"

Step 3: Bedside tests

Test	How	What it tells you
Weight (and serial weights)	Same scale, undressed	Exact fluid volumes; weight gain confirms rehydration (1 g ≈ 1 mL)
Random blood sugar	Glucometer	Hypoglycaemia (< 3 mmol/L) in lethargic, malnourished or vomiting children
Malaria RDT	Finger prick	Malaria presenting with vomiting and diarrhoea
Look at the stool	Ask to see the nappy	Blood, mucus, rice-water stool, steatorrhoea
Urine dipstick and output	Bag or pad	Specific gravity, ketones, glucose (DKA!), UTI; first urine = rehydration working
HemoCue Hb	Finger prick	Severe anaemia changes fluid choices; HUS
HIV rapid test (if status unknown)	Finger prick, after counselling	Persistent diarrhoea, malnutrition, recurrent illness
Cholera RDT (outbreak setting)	Stool dipstick	Screens suspected cholera; confirm with culture

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
Serum electrolytes, urea, creatinine	Severe dehydration, IV fluids, convulsions, doughy skin, SAM, persistent vomiting	Na < 130 or > 150; K < 3.5; raised urea and creatinine (pre-renal AKI); low bicarbonate	Slow correction of hypernatraemia; add potassium once passing urine; renal review. **Regional/national referral hospitals**; rarely at HC IV
Stool microscopy	Persistent diarrhoea, dysentery not responding, HIV	Trophozoites of *E. histolytica* with ingested RBCs; *Giardia* cysts or trophozoites; ova (*Strongyloides*, *S. mansoni*); *Cryptosporidium* (modified ZN)	Metronidazole only when amoeba or giardia is **seen**. HC III and above
Stool culture and sensitivity	Dysentery not improving after 2 days, outbreak, suspected cholera	*Shigella*, *Salmonella*, *V. cholerae* with sensitivity	Change antibiotic; public health action. **RRH, Mulago, CPHL**
Full blood count	Severe illness, dysentery, pallor	Leucocytosis, anaemia, low platelets (HUS)	Transfusion; HUS work-up. Most HC IVs
Blood culture	Sepsis, SAM, infant < 2 months, typhoid suspected	Gram-negative bacteraemia	IV antibiotics. RRH/NRH

Test	When	Expected / abnormal finding	How it changes management
Blood film/smear	HUS suspected	Fragmented red cells (schistocytes)	Referral for dialysis
Abdominal ultrasound	Bloody stool with colicky pain, mass, bilious vomiting	Target sign (intussusception); free fluid	Surgical referral; air/saline reduction at NRH
Abdominal X-ray	Distension, bilious vomiting	Fluid levels (obstruction), pneumatosis (NEC)	Surgery; NG drainage

Step 5: Putting it together

Model one-line summary

> "A 10-month-old girl, 7.0 kg (pre-illness 7.6 kg on her card), with 2 days of acute watery diarrhoea (8 stools/day) and vomiting, who is restless, has sunken eyes, drinks eagerly and has a slow skin pinch: acute watery diarrhoea with **some dehydration**, no blood, no SAM (MUAC 13.0 cm, no oedema), HIV-unexposed and fully immunised for age."

Problem list

1. Acute watery diarrhoea with some dehydration (about 8% from the weight loss).
2. Vomiting (not bilious).
3. Fever 38.2 °C: malaria RDT pending.
4. Feeding: breastfed plus porridge; continue.

Working diagnosis: *acute gastroenteritis (probably viral) with some dehydration.* **Plan B**: ORS 75 × 7.0 = **525 mL over 4 hours**, zinc 20 mg daily for 10 days, continue breastfeeding.

EXAM FAVOURITE

Worked example 1: how much dehydration from the weights? Pre-illness weight 7.6 kg, today 7.0 kg. Loss = 0.6 kg ≈ **600 mL**. Percentage = 0.6 ÷ 7.6 = **7.9%** = "some dehydration" (5–10%). Plan B replaces 75 mL/kg × 7.0 = **525 mL**, which is close to the measured loss. After 4 hours she weighs 7.5 kg, is alert, drinks normally and the skin pinch is normal: reclassify as **no dehydration → Plan A**.

EXAM FAVOURITE

Worked example 2: Plan C, step by step. A 2-year-old boy, **11 kg**, is lethargic, cannot drink, eyes very sunken, skin pinch 3 s, cold hands, CRT 4 s. MUAC 14 cm, no oedema. **Severe dehydration (with shock), no SAM.** He is **≥ 12 months**, so Ringer's lactate **100 mL/kg = 1100 mL**:

- **30 mL/kg = 330 mL in 30 minutes** (with a 20-drop/mL giving set that is 330 × 20 ÷ 30 = **220 drops/min**: practically "run it as fast as it will go", squeezing the bag);
- reassess: radial pulse still very weak → **repeat 330 mL once**;
- then **70 mL/kg = 770 mL over 2.5 hours** = about **308 mL/h** (about 100 drops/min with a 20-drop set; 1 drop every 0.6 s);
- **ORS 5 mL/kg/h (55 mL/h)** as soon as he can drink;
- reassess every 15–30 min; full reclassification at **3 hours** (6 hours in an infant).

UGANDA CONTEXT

UCG 2023 says reassess **every 30–60 minutes** during Plan C; the WHO Pocket Book says every **15–30 minutes**. In practice: check pulse, CRT and conscious level **every 15 minutes during the first fast phase**, then every 30–60 minutes. A busy ward will not do this unless you write it in the plan.

Differential diagnosis

Differentials of **a child with vomiting and/or diarrhoea**:

Condition	Distinguishing features	Key investigation
Acute viral gastroenteritis (rotavirus, norovirus)	Vomiting first, then watery diarrhoea; low-grade fever; under 2 years	Clinical
Cholera	Profuse painless "rice-water" stools, rapid severe dehydration, often older children too; outbreak	Cholera RDT, stool culture
Bacillary dysentery (*Shigella*)	Blood and mucus, fever, tenesmus, abdominal cramps; may convulse	Stool microscopy (pus cells), culture
Amoebic dysentery	Bloody mucoid stools, less fever, older children; rare under 5 years	Trophozoites with ingested RBCs
Malaria	Fever, vomiting, loose stools, pallor, splenomegaly	RDT / blood smear
UTI / pyelonephritis	Fever and vomiting in an infant, foul urine	Urine dipstick, microscopy, culture
Pneumonia, otitis media	Fast breathing, chest indrawing; ear discharge	Clinical, SpO_2
Meningitis	Vomiting, fever, convulsions, bulging fontanelle, neck stiffness	Lumbar puncture
Intussusception	3 months–3 years, episodic screaming and pallor, red-currant-jelly stool, sausage mass	Ultrasound (target sign)
Intestinal obstruction / volvulus	**Bilious vomiting**, distension, absolute constipation	X-ray, surgical review
Appendicitis	Older child, central then RIF pain, fever, guarding; diarrhoea in pelvic appendix	Clinical, ultrasound
Diabetic ketoacidosis	Vomiting, abdominal pain, **deep breathing**, polyuria despite "dehydration"	RBS, urine glucose and ketones
Sepsis	Very ill, fever or hypothermia, mottled; may have diarrhoea	Blood culture, FBC
Haemolytic uraemic syndrome	5–10 days after bloody diarrhoea: pallor, oliguria, petechiae	FBC, film, creatinine
Antibiotic-associated, drugs, herbal purgatives	Temporal link to medicines or local herbs	History
Persistent diarrhoea causes (HIV enteropathy, giardia, cryptosporidium, lactose intolerance, SAM)	≥ 14 days, weight loss	HIV test, stool microscopy, nutritional assessment

Investigations

Test	What you are looking for	Why
Weight (serial)	Deficit and response	Volumes are per kg; weight gain shows rehydration
RBS	Hypoglycaemia	Treatable cause of lethargy and convulsions
Malaria RDT	*P. falciparum*	Common co-diagnosis in febrile children
Electrolytes, urea, creatinine	Na, K, AKI	Severe cases, IV fluids, convulsions, SAM
Stool microscopy and culture	Pathogen, parasites	Dysentery not improving, persistent diarrhoea, outbreaks
FBC	Anaemia, infection, thrombocytopenia	Severe illness, HUS
HIV test	HIV infection	Persistent diarrhoea, SAM, recurrent illness
Urinalysis	UTI, glucose/ketones	Alternative diagnoses

Most children with acute watery diarrhoea need **no laboratory test at all**: the diagnosis and plan come from the bedside assessment.

Management

A. Principles (the five WHO pillars)

1. **Rehydrate** with the right plan (A, B or C), using **low-osmolarity ORS** (WHO 2024, strong recommendation, now for children up to 10 years).
2. **Zinc** for every child with diarrhoea.
3. **Continue feeding** (and breastfeeding), with extra food during recovery.
4. **Antibiotics only when indicated**: dysentery, cholera with dehydration, or another bacterial infection (WHO 2024 suggests **against** antibiotics in acute watery diarrhoea).
5. **Counsel the mother**: home fluids, danger signs and when to return, hygiene and vaccines.

B. Plan A: no dehydration (treat at home)

The 4 rules of home treatment:

1. **Give extra fluid** (as much as the child will take):
   - Breastfeed more often and for longer at each feed.
   - Exclusively breastfed baby: give **ORS or clean water in addition** to breast milk.
   - Otherwise: ORS, soup, rice water, yoghurt drinks, clean water.
   - **ORS after each loose stool**: **under 2 years 50–100 mL**; **2 years and older 100–200 mL** (older children as much as they want).
   - Give **2 packets of ORS** to take home; teach how to mix **one packet in 1 litre** of clean water; discard after 24 h.
   - Small, frequent sips from a **cup** (spoon for babies). If the child vomits, **wait 10 minutes**, then continue more slowly.
2. **Give zinc** (dose table below) for 10 days, even after the diarrhoea stops.
3. **Continue feeding**: breastfeed; give usual foods, fresh and well cooked, in small frequent meals; an **extra meal a day for 2 weeks** after the episode.

4. **Know when to return immediately**: not able to drink or breastfeed, becomes sicker, develops fever, **blood in the stool**, drinking poorly. Follow up in **5 days** if not improving.

C. Plan B: some dehydration (ORS in the clinic)

Amount of ORS in the first 4 hours = weight (kg) × 75 mL. Use age only if you cannot weigh.

Age	Up to 4 months	4–12 months	12 months–2 years	2–5 years
Weight	< 6 kg	6–< 10 kg	10–< 12 kg	12–19 kg
ORS in 4 h	**200–400 mL**	**400–700 mL**	**700–900 mL**	**900–1400 mL**

1. Sit the mother down in the ORT corner with a **measured** amount in a jug and a cup and spoon.
2. Give **frequent small sips**; for an infant, **1 teaspoon every 1–2 minutes**. If the child wants more, give more.
3. Continue **breastfeeding**. A non-breastfed infant under 6 months also gets **100–200 mL of clean water** during these 4 hours.
4. If vomiting, **wait 10 minutes** and continue slowly. If vomiting persists or the child cannot drink, use an **NG tube** at the same rate.
5. **Reassess every 30–60 minutes**; if signs of severe dehydration appear, move to Plan C.
6. **After 4 hours**: reassess and reclassify → Plan A, B (repeat) or C. **Start feeding in the clinic.**
7. Give **zinc** and, if the mother must leave early, enough ORS to finish the 4-hour amount plus 2 packets for Plan A.

D. Plan C: severe dehydration (rapid IV rehydration)

If you can give IV fluids (HC III and above):

1. Start IV (or IO) immediately; if the child can drink, give **ORS by mouth while the drip is set up**.
2. Give **Ringer's lactate 100 mL/kg** (if not available: normal saline 0.9%; half-strength Darrow's in 2.5% glucose is an older alternative). **Never plain 5% dextrose**: it does not rehydrate.

Age	First give 30 mL/kg in	Then give 70 mL/kg in
Infants under 12 months	**1 hour**	**5 hours**
Children 12 months to 5 years	**30 minutes**	**2.5 hours**

Repeat the first 30 mL/kg once if the radial pulse is still very weak or not detectable.

3. **Reassess** every 15–30 minutes (UCG: 30–60 minutes): if hydration is not improving, give the drip faster.
4. Give **ORS about 5 mL/kg/h** as soon as the child can drink (usually after **3–4 hours in infants, 1–2 hours in children**).
5. **Reclassify** after 6 hours (infant) or 3 hours (child), and choose Plan A, B or C to continue.
6. Observe for **at least 6 hours** after rehydration to make sure the mother can keep the child hydrated with ORS.

If IV fluids are not available:

- IV treatment **within 30 minutes**: refer urgently; mother gives sips of ORS on the way.
- **No IV nearby but an NG tube or the child can drink**: **ORS 20 mL/kg/h for 6 hours (total 120 mL/kg)** by NG tube or mouth; reassess every 1–2 hours; slow down if repeated vomiting or abdominal distension; **refer if not improving after 3 hours**.

RED FLAG

Shock (cold hands, CRT > 3 s, weak fast pulse) with severe dehydration: the first 30 mL/kg of Plan C **is** the resuscitation, given as fast as possible; UCG 2023's shock protocol gives **20 mL/kg as rapidly as possible**, repeated if needed, then **70 mL/kg over 5 h (infant) or 2.5 h (child)**. Check **glucose** and look for **severe anaemia** (Hb < 5 g/dL → blood, not more crystalloid). **Stop** and reassess if crackles, a gallop rhythm, a liver that is enlarging or puffy eyelids appear.

NOTE

WHO ETAT 2016 and the FEAST trial: children who are febrile and have **impaired circulation without full shock** (only 1–2 of the 3 signs) should **not** get rapid boluses; give maintenance fluids and reassess within 1 hour. Children in **full shock** get **10–20 mL/kg over 30–60 min**. Severe **dehydration from diarrhoea** is the main exception where rapid fluid (Plan C) is still correct, because the child has measurable losses to replace.

E. Drugs

Drug	Dose (mg/kg)	Route	Frequency	Duration	Max / notes
Low-osmolarity ORS	Plan A: 50–100 mL (< 2 y) or 100–200 mL (≥ 2 y) per loose stool. Plan B: **75 mL/kg** over 4 h. NG Plan C: **20 mL/kg/h**	Oral / NG	Continuous sips	Until diarrhoea stops	1 sachet in 1 L of clean water; discard after 24 h
Zinc sulphate (dispersible 20 mg tab)	**< 6 months: 10 mg** (½ tab); **≥ 6 months: 20 mg** (1 tab)	Oral, dissolved in breast milk or water	Once daily	**10 days** (UCG 2023); WHO: 10–14 days	**WHO 2024 suggests 5 mg daily** (less vomiting); **not needed if on RUTF/F-75/F-100** (already contain zinc)
Ciprofloxacin (dysentery, first line)	**15 mg/kg**	Oral	**12-hourly**	**3 days**	Max 500 mg/dose. 500 mg tab: under 6 months ¼ tab; 6 months–5 years ½ tab (UCG)
Ceftriaxone (dysentery, severely ill or treatment failure)	**50–80 mg/kg**	IV or IM	Once daily	**3 days**	Max 2 g; **not with calcium-containing IV fluids** in neonates
Doxycycline (cholera, Uganda MoH 2025, children < 12 y)	**2–4 mg/kg**	Oral	**Single dose**	Once	Max 300 mg. Repeat if vomited within 30 min
Azithromycin (cholera alternative; dysentery alternative per local sensitivity)	**20 mg/kg**	Oral	**Single dose** (cholera)	Once	Max 1 g
Erythromycin (cholera, older UCG IMCI option)	125 mg (< 2 y); 250 mg (2–5 y)	Oral	6-hourly	3 days	Use where doxycycline/azithromycin are unavailable
Metronidazole (amoebiasis **seen** on microscopy)	**10 mg/kg**	Oral	**8-hourly**	**5–10 days** (UCG adult course 10 days)	Max 800 mg/dose. Only if trophozoites with RBCs seen, or dysentery fails 2 antibiotics

Drug	Dose (mg/kg)	Route	Frequency	Duration	Max / notes
Metronidazole (giardiasis)	**5–7.5 mg/kg**	Oral	8-hourly	5–7 days	WHO 5 mg/kg for 5 days; UCG (SAM) 7.5 mg/kg for 7 days
Vitamin A (persistent diarrhoea, measles, SAM with eye signs)	**< 6 months 50,000 IU; 6–11 months 100,000 IU; ≥ 12 months 200,000 IU**	Oral	Single dose (days 1, 2 and 14 if eye signs)		Skip if a dose was given in the last month
10% glucose (hypoglycaemia)	**5 mL/kg**	IV	Once, recheck in 30 min		Or 50 mL sugar water oral/NG

DRUG DOSING

Do not give: antidiarrhoeals (loperamide, codeine, kaolin), antiemetics (chlorpromazine/Largactil, promethazine, metoclopramide) in young children, **metronidazole "for diarrhoea"**, or **probiotics** (WHO 2024 suggests against). They do not save lives and some (antimotility drugs, chlorpromazine) are dangerous. Ondansetron is used in some high-income emergency departments for vomiting, but it is not part of the WHO or Ugandan protocols.

UGANDA CONTEXT

Zinc dose: Uganda vs WHO 2024. UCG 2023 (both the IMCI chart and the diarrhoea chapter) still uses **10 mg (< 6 months) / 20 mg (≥ 6 months) for 10 days**. The **WHO 2024 Guideline on management of pneumonia and diarrhoea in children up to 10 years** now *suggests* **5 mg daily for 10–14 days**, because it works as well with about half the vomiting. Until the MoH updates the guideline and the dispersible tablet, Ugandan wards and HC dispensaries give the 10/20 mg tablet. In an exam, give the UCG dose and say WHO now suggests 5 mg.

F. Dysentery (blood in the stool)

1. **Assess dehydration** and treat with the right plan.
2. **Ciprofloxacin 15 mg/kg 12-hourly for 3 days** (UCG 2023, WHO 2024 remarks). If **severely ill**, under 2 months, or not better after 2 days: **ceftriaxone 50–80 mg/kg IV/IM daily for 3 days** and admit.
3. **Review in 2–3 days** (UCG: follow up in 3 days): better = fewer stools, less blood, less fever, eating better. If not, switch to the second-line drug, send stool for culture and microscopy, and look for **amoebae**.
4. **Young infants (< 2 months) with blood in the stool**: refer; consider **intussusception** and **surgical causes**.
5. Zinc, feeding, paracetamol for fever and pain. **No antimotility drugs** (risk of toxic megacolon).
6. Watch for complications: **convulsions** (Shigella neurotoxin, fever, hyponatraemia, hypoglycaemia), **rectal prolapse** (push back gently with a gloved hand), **HUS** (oliguria, pallor, low platelets), and **intestinal perforation**.

G. Persistent diarrhoea (≥ 14 days)

Severe persistent diarrhoea (with dehydration) or any child < 4 months: admit.

1. **Rehydrate** (Plans A–C); watch electrolytes.
2. **Look for and treat infections**: **HIV test**, stool microscopy (giardia, amoebae, cryptosporidium, *Strongyloides*), UTI, pneumonia, otitis media, TB. **Antibiotics only for a specific infection**; metronidazole only when giardia or amoebae are found.
3. **Feeding** is the treatment:

- Continue breastfeeding, more often and for longer.
- **Reduce animal milk to no more than 50 mL/kg/day**, or replace it with **yoghurt** or fermented milk (less lactose).
- Give energy-dense family foods (porridge with oil and groundnut paste, mashed beans and vegetables) **6 times a day**.
- True **lactose intolerance** (diarrhoea restarts every time milk is reintroduced) is uncommon; use a reduced-lactose diet then reintroduce.

4. **Micronutrients for 14 days**: **multivitamins and minerals including zinc** (UCG IMCI); **vitamin A** single dose (100,000 IU at 6–11 months; 200,000 IU at 1 year or more); folate.
5. **Follow up in 5 days** (outpatients); weigh daily as an inpatient: the diet is working when the child **gains weight** and the stools fall to fewer than 3 a day.

H. Cholera

1. **Notify** (it is a notifiable disease; call the District Surveillance Focal Person). Admit to a **cholera treatment unit (CTU)**; isolate; chlorine (0.05% hand-washing, 0.5% for surfaces and stool).
2. **Rehydrate** by the same plans: **Plan B 75 mL/kg over 4 h**; **Plan C Ringer's lactate 100 mL/kg** (30 mL/kg fast then 70 mL/kg) and **repeat once if still severely dehydrated** at reassessment. Losses can be enormous: **replace ongoing losses volume for volume** with ORS until the diarrhoea stops.
3. **Antibiotics for all cholera patients** (Uganda MoH Cholera Guidelines 2025): children < 12 years **doxycycline 2–4 mg/kg single dose**; alternatives **azithromycin 20 mg/kg single dose (max 1 g)** or ciprofloxacin per the guideline. If vomited within 30 minutes, repeat the dose.
4. **Zinc** for children under 5 (not if on RUTF).
5. **SAM with cholera** (Uganda 2025): use **standard ORS, not ReSoMal** (cholera stools are very salty); monitor closely for overload; severe dehydration without shock should not get IV fluids.
6. **Do not give** metronidazole, chlorpromazine or loperamide.
7. Prevention: oral cholera vaccine in outbreaks, safe water, latrines, hand-washing; chemoprophylaxis for household contacts per the guideline.

I. When severe acute malnutrition changes the plan

Feature	Well-nourished child	Child with SAM
Diagnosing dehydration	Four signs	Signs unreliable (sunken eyes and slow pinch occur anyway); rely on **history of recent watery stools, recent change in appearance, thirst, and weight**
Fluid	Low-osmolarity ORS	**ReSoMal** (Na 45, K 40 mmol/L); if not available, low-osmolarity ORS (WHO 2023) or half-strength ORS with added glucose and potassium (UCG). In **cholera: standard ORS**
Some/severe dehydration, not in shock	Plan B or C	**No IV.** ReSoMal **5 mL/kg every 30 min for 2 h**, then **5–10 mL/kg/h for the next 4–10 h** (alternating with F-75 in the same volume)
Shock	Plan C / 20 mL/kg	**15 mL/kg over 1 hour** of Ringer's lactate with 5% glucose, or half-strength Darrow's with 5% glucose, or 0.45% saline with 5% glucose; **repeat once** if improving, then ReSoMal by mouth/NG; **if not improving: blood 10 mL/kg over 3 h** and treat for septic shock
Monitoring	Hydration signs	**Every 30 min for 2 h, then hourly**: pulse, RR, liver size, urine, stool. **Stop if pulse rises by 25/min and RR by 5/min** (overload)
Ongoing losses	ORS per stool	ReSoMal **50–100 mL per watery stool** (UCG); small frequent unformed stools in SAM need no treatment

Feature	Well-nourished child	Child with SAM
Zinc	Zinc tablets	Already in F-75, F-100 and RUTF; no extra tablets
Antibiotics	Only if indicated	**All** SAM children get antibiotics (see Malnutrition topic)

J. Setting-specific plan

Setting	What you can do
Village health team / HC II	Classify; Plans A and B; zinc; pre-referral ORS sips; refer severe cases, dysentery in young infants, SAM with complications
HC III	Plan C with IV Ringer's lactate; NG ORS; ciprofloxacin; RBS and malaria RDT; refer if not improving
HC IV	Inpatient care, IV antibiotics (ceftriaxone), stool microscopy, Hb, blood transfusion (if blood fridge), admission of severe cases
Regional / national referral (Mengo, Mulago, RRH)	Electrolytes, stool culture, ultrasound, surgery for intussusception, ICU care, specialist SAM wards (e.g. Mwanamugimu Nutrition Unit, Mulago)

K. Discharge and prevention

- Discharge when **rehydrated, feeding, passing urine**, the mother has shown she can give ORS, and any danger signs have resolved.
- Counsel: **4 rules**, **hand-washing with soap** (after the latrine, before preparing food), **safe water** (boil or treat), **latrine use**, **exclusive breastfeeding to 6 months**, **rotavirus and measles–rubella vaccines**, **vitamin A** every 6 months from 6 months to 5 years.

Complications

Group	Complications
Fluid	**Hypovolaemic shock**, **pre-renal AKI → acute tubular necrosis**, death; **fluid overload** and heart failure from over-rapid IV fluids (especially SAM, severe anaemia)
Electrolytes and metabolic	**Hypo- and hypernatraemia** (convulsions; cerebral oedema if hypernatraemia is corrected too fast), **hypokalaemia** (ileus, weakness, arrhythmia), **metabolic acidosis**, **hypoglycaemia**
Infective	**Sepsis**, bacteraemia (non-typhoidal *Salmonella* in HIV, malaria and SAM), **HUS** (*S. dysenteriae* type 1, EHEC), reactive arthritis, toxic megacolon, intestinal perforation
Local	**Rectal prolapse** (*Shigella*, trichuriasis), perianal excoriation, intussusception
Nutritional	Weight loss, **persistent diarrhoea**, **malnutrition**, stunting with repeated episodes, zinc deficiency, transient lactose intolerance
Neurological	Convulsions (fever, sodium, glucose, *Shigella* toxin), cerebral venous thrombosis in severe hypernatraemic dehydration

Ward-round & exam pearls

Model case presentation:

> "Baby A is a **10-month-old girl** from Kisenyi, brought by her mother with **2 days of watery, non-bloody diarrhoea about 8 times a day** and vomiting 4 times a day. She is breastfed and eats posho and beans. She has had **low-grade fever**, no convulsions, and passed urine 6 hours ago. She was **born at term, 3.1 kg**, is **up to date on UNEPI** for age including **both rotavirus doses**, and is **HIV-unexposed** (mother negative in this pregnancy and at 6 weeks postnatal). Development is normal. On examination she is **restless and irritable**, eyes sunken, **drinks eagerly**, skin pinch goes back slowly. **Weight 7.0 kg** (7.6 kg two weeks ago at the immunisation clinic), **length 70 cm**, **WLZ –1** and **WAZ –1.4**, **MUAC 13.0 cm**, **no oedema**. Temp 38.2 °C, HR 150, RR 42, CRT 2 s, hands warm. The abdomen is soft with no masses. Malaria RDT negative, RBS 4.6 mmol/L. **Summary**: acute watery diarrhoea with **some dehydration** (about 8%), no SAM. **Plan B**: ORS 525 mL over 4 hours in the ORT corner, zinc 20 mg once daily for 10 days, continue breastfeeding; reassess at 4 hours."

Likely viva questions (with answers)

1. *How do you classify dehydration?* Two or more signs: severe = lethargic/unconscious, sunken eyes, unable to drink, skin pinch very slow; some = restless/irritable, sunken eyes, thirsty, skin pinch slow; otherwise none.
2. *What is in low-osmolarity ORS and why is it better?* Na 75, glucose 75, K 20, Cl 65, citrate 10 mmol/L; 245 mOsm/L; less stool output, less vomiting and fewer IV drips than the old 311 mOsm/L formula.
3. *Give the Plan C volumes for a 9-month-old weighing 8 kg.* Ringer's lactate 800 mL: 240 mL in the first hour, then 560 mL over 5 hours; ORS 40 mL/h once drinking; reassess at 6 h.
4. *When do you give antibiotics in diarrhoea?* Dysentery, cholera, and co-existing bacterial infections (pneumonia, UTI, sepsis), plus all children with SAM. Not for watery diarrhoea.
5. *Why zinc?* Shortens duration and severity and prevents further episodes for 2–3 months.
6. *How does SAM change rehydration?* No IV unless shocked; ReSoMal slowly (5 mL/kg every 30 min for 2 h, then 5–10 mL/kg/h); shock 15 mL/kg over 1 h; watch for overload.
7. *What is the danger of correcting hypernatraemia quickly?* Cerebral oedema and convulsions; lower sodium by no more than 0.5 mmol/L/h (10–12 mmol/L in 24 h), usually over 48 h or more.

OSCE tips: in the "assess and classify a child with diarrhoea" station, **wash hands, greet the mother, check general danger signs first**, then ask about duration and blood, look at general condition and eyes, **offer a drink**, do the skin pinch correctly, and **say the classification and plan out loud** with the exact volume. For "counsel a mother on ORS", demonstrate mixing **1 sachet in 1 litre**, the amount after each stool, sips by cup and spoon, and the **3 return signs**.

Mnemonics & memory aids

- **"Two signs make the class"**: you need **2 or more** signs to classify as some or severe.
- **Plan C "30–70, 1–5 or ½–2½"**: 30 mL/kg then 70 mL/kg; infants over **1 h + 5 h**, children over **½ h + 2½ h**.
- **Plan B "75 in 4"**: 75 mL/kg in 4 hours.
- **NG rescue "20 for 6"**: 20 mL/kg/h for 6 hours.
- **ReSoMal "5 every 30 for 2, then 5–10 an hour"**.
- **The 4 rules of Plan A: "Fluids, Feed, Zinc, Return" (FFZR)**.
- **Dysentery "Cipro 15 × 2 × 3"**: 15 mg/kg, twice daily, 3 days.
- **SAM shock "15 in 1"**: 15 mL/kg in 1 hour.

High-yield summary

Numbers & doses to know

Item	Value
Diarrhoea	≥ **3 loose/watery stools in 24 h**
Persistent diarrhoea	≥ **14 days**
Degree of dehydration	None < 5%; some 5–10%; severe > 10%
Skin pinch	Slow < 2 s (some); **very slow > 2 s** (severe)
Plan A ORS per stool	**50–100 mL** (< 2 y); **100–200 mL** (≥ 2 y)
Plan B	**75 mL/kg over 4 h**
Plan C, < 12 months	**30 mL/kg in 1 h, then 70 mL/kg in 5 h** (RL)
Plan C, ≥ 12 months	**30 mL/kg in 30 min, then 70 mL/kg in 2.5 h**
NG ORS if no IV	**20 mL/kg/h × 6 h (120 mL/kg)**
ORS once drinking in Plan C	**5 mL/kg/h**
Low-osmolarity ORS	Na **75**, glucose **75**, K **20**, Cl **65**, citrate **10** mmol/L; **245 mOsm/L**
ReSoMal	Na **45**, K **40** mmol/L
Zinc (UCG 2023)	**10 mg** (< 6 mo) / **20 mg** (≥ 6 mo) daily × **10 days**; WHO 2024 suggests **5 mg** for 10–14 days
Dysentery	**Ciprofloxacin 15 mg/kg 12-hourly × 3 days**; ceftriaxone **50–80 mg/kg daily × 3 days**
Cholera (child)	**Doxycycline 2–4 mg/kg single dose** or azithromycin **20 mg/kg** single dose (max 1 g)
SAM dehydration	ReSoMal **5 mL/kg every 30 min × 2 h**, then **5–10 mL/kg/h** up to 10 h
SAM shock	**15 mL/kg over 1 h** (glucose-containing fluid); blood **10 mL/kg over 3 h** if not improving
Hypoglycaemia	RBS < 3 mmol/L → **10% glucose 5 mL/kg IV**
Fast breathing	≥ 60 (< 2 mo), ≥ 50 (2–12 mo), ≥ 40 (1–5 y)

Classic exam traps

- **"Severe dehydration in a marasmic child → Plan C."** Wrong: no IV unless in shock; ReSoMal slowly.
- **"Give metronidazole for diarrhoea."** Wrong: only for amoebae or giardia seen on microscopy.
- **"Stop breastfeeding to rest the gut."** Wrong: breastfeeding continues throughout, including during Plan B and C.
- **"Child is vomiting, so ORS won't work, start a drip."** Wrong: some dehydration with vomiting = small sips or NG ORS.
- **"5% dextrose for rehydration."** Wrong: it contains no sodium; use Ringer's lactate or normal saline.
- **"Number of stools tells you the dehydration."** Wrong: grade with the signs.
- **"Antibiotics shorten watery diarrhoea."** Wrong: WHO 2024 suggests against antibiotics in acute watery diarrhoea; give them for blood or cholera.
- **"Zinc stops when the diarrhoea stops."** Wrong: complete the full 10 days.

Questions seniors ask

This child has diarrhoea and is very wasted. How will you rehydrate? Confirm SAM (MUAC, WLZ, oedema). If not shocked: ReSoMal 5 mL/kg every 30 min for 2 h, then 5–10 mL/kg/h, with F-75, monitoring pulse, RR, liver and urine every 30–60 min. If shocked: 15 mL/kg over 1 h of RL with 5% glucose, then oral/NG ReSoMal or blood

if not improving.

How do you know Plan B worked? Reassess after 4 hours: the child is alert, drinks normally, eyes less sunken (ask the mother), skin pinch normal, passing urine, and has gained weight. Then Plan A.

What would make you think this is not gastroenteritis? Bilious vomiting, a distended or tender abdomen, a mass, blood with episodic screaming, high fever with neck stiffness, deep breathing with glycosuria, or vomiting without diarrhoea.

Why Ringer's lactate rather than normal saline? It contains potassium, lactate (a bicarbonate source) and less chloride, so it corrects acidosis better; normal saline is acceptable if RL is unavailable.

When would you check electrolytes? Severe dehydration needing IV fluids, convulsions or altered consciousness, doughy skin or hypernatraemia suspected, SAM, persistent vomiting, or no urine after rehydration.

What does the WHO 2024 guideline change? It extends the recommendations to children up to 10 years: low-osmolarity ORS (strong), zinc for acute and persistent diarrhoea with a suggested 5 mg dose, no antibiotics for watery diarrhoea, antibiotics for bloody diarrhoea, and no probiotics.

A child with dysentery is no better after 2 days of ciprofloxacin. Next step? Admit if unwell, send stool for microscopy and culture, switch to ceftriaxone 50–80 mg/kg daily, look for amoebae (metronidazole if trophozoites seen) and complications (HUS, intussusception, dehydration, hypoglycaemia).

SELF-TEST

Flashcards

28 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Definition of diarrhoea (WHO)	**3 or more loose or watery stools in 24 hours**. Frequent pasty stools in a breastfed baby are not diarrhoea.
2	Persistent diarrhoea vs dysentery	**Persistent**: an episode lasting **14 days or more**. **Dysentery**: diarrhoea with **visible blood** in the stool (any duration).
3	Signs of SEVERE dehydration (need 2 or more)	**Lethargic or unconscious**; sunken eyes; **not able to drink or drinks poorly**; skin pinch goes back **very slowly (> 2 s)**.
4	Signs of SOME dehydration (need 2 or more)	**Restless, irritable**; sunken eyes; **drinks eagerly, thirsty**; skin pinch goes back **slowly**.
5	The 4 rules of Plan A (home treatment)	1. **Extra fluid** (ORS after each loose stool) 2. **Zinc** 3. **Continue feeding** 4. **When to return** (cannot drink, sicker, fever, blood in stool)
6	Plan A: ORS after each loose stool	Under 2 years: **50–100 mL**. 2 years and older: **100–200 mL**. Give **2 packets** to take home.
7	Plan B volume and time	**ORS 75 mL/kg over 4 hours** in the clinic, then reassess and reclassify.
8	Plan B by age (no scale): ORS in the first 4 hours	< 4 mo (< 6 kg) **200–400 mL**; 4–12 mo (6–< 10 kg) **400–700 mL**; 12 mo–2 y (10–< 12 kg) **700–900 mL**; 2–5 y (12–19 kg) **900–1400 mL**.
9	Plan C in an infant under 12 months	Ringer's lactate **100 mL/kg**: **30 mL/kg in 1 hour**, then **70 mL/kg in 5 hours**. Reassess at 6 hours.
10	Plan C in a child 12 months to 5 years	Ringer's lactate **100 mL/kg**: **30 mL/kg in 30 minutes**, then **70 mL/kg in 2.5 hours**. Reassess at 3 hours.
11	Plan C when no IV is available	ORS by **NG tube (or mouth) 20 mL/kg/h for 6 hours** (120 mL/kg); reassess every 1–2 h; **refer if not improving after 3 hours**.
12	When do you repeat the first 30 mL/kg in Plan C?	Once, if the **radial pulse is still very weak or not detectable** after the first 30 mL/kg.
13	ORS during Plan C once the child can drink	**About 5 mL/kg/h** (usually after 3–4 h in infants, 1–2 h in children).
14	Composition of low-osmolarity ORS	Na **75**, glucose **75**, K **20**, Cl **65**, citrate **10** mmol/L; total **245 mOsm/L**.
15	Zinc dose (UCG 2023)	Under 6 months **10 mg**; 6 months or older **20 mg**, once daily for **10 days**. WHO 2024 suggests **5 mg daily for 10–14 days**.
16	Why does zinc help in diarrhoea?	Replaces zinc lost in stool; speeds mucosal repair and absorption; **shortens the episode, reduces stool volume** and reduces diarrhoea over the next 2–3 months.
17	First-line antibiotic for dysentery (Uganda/WHO)	**Ciprofloxacin 15 mg/kg orally 12-hourly for 3 days**.
18	Second-line / severely ill dysentery	**Ceftriaxone 50–80 mg/kg IV or IM once daily for 3 days**; admit.
19	Antibiotic for cholera in a child under 12 (Uganda MoH 2025)	**Doxycycline 2–4 mg/kg orally, single dose**. Alternative: **azithromycin 20 mg/kg single dose (max 1 g)**.
20	Why does ORS work even in cholera?	**Sodium–glucose co-transport (SGLT1)** in villous cells is intact; glucose carries sodium in, and water follows sodium.

#	Question	Answer
21	Rehydrating a child with SAM who is NOT shocked	**No IV.** ReSoMal **5 mL/kg every 30 min for 2 h**, then **5–10 mL/kg/h for up to 10 h** (alternate with F-75).
22	Shock in a child with SAM: IV fluid	**15 mL/kg over 1 hour** of RL with 5% glucose (or half-strength Darrow's or 0.45% saline with 5% glucose); repeat once if improving; if not, **blood 10 mL/kg over 3 h**.
23	Signs of fluid overload during rehydration	**Pulse up by 25/min and RR up by 5/min**, crackles, enlarging liver, gallop, puffy eyelids, raised JVP. **Stop fluids and reassess.**
24	Feeding in persistent diarrhoea	Continue breastfeeding; **animal milk ≤ 50 mL/kg/day** or replace with **yoghurt**; energy-dense meals 6 times a day; **multivitamins + minerals (with zinc) for 14 days**; vitamin A.
25	Drugs NOT to give in childhood diarrhoea	Antimotility drugs (loperamide), antiemetics like chlorpromazine, **metronidazole without a parasite seen**, probiotics (WHO 2024), and **plain 5% dextrose** for rehydration.
26	Red-currant jelly stool + episodic screaming in a 7-month-old	**Intussusception** until proven otherwise: ultrasound (target sign), surgical referral; do not label as dysentery.
27	Hypernatraemic dehydration: clue and danger	**Doughy skin**, marked irritability and thirst; signs underestimate the deficit. Correct slowly: Na fall **≤ 0.5 mmol/L/h** (≤ 10–12 mmol/L in 24 h) or **cerebral oedema** and convulsions follow.
28	Shock (WHO ETAT 2016) = all three of …	**Cold extremities + capillary refill > 3 s + weak, fast pulse.** One or two signs alone = impaired circulation, not shock.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 14-month-old boy (9.5 kg) has had watery diarrhoea for 2 days. He is restless and irritable, his eyes are sunken, he grabs the cup and drinks eagerly, and the skin pinch goes back slowly. CRT 2 s, hands warm. MUAC 13.5 cm, no oedema. What is the MOST appropriate management?

- A ORS about 700 mL over 4 hours in the clinic, zinc 20 mg daily, continue feeding, then reassess
- B Ringer's lactate 285 mL over 30 minutes, then 665 mL over 2.5 hours
- C Two ORS packets for home, zinc 20 mg daily and review in 5 days
- D ReSoMal 5 mL/kg every 30 minutes for 2 hours
- E Ciprofloxacin 15 mg/kg 12-hourly for 3 days and ORS at home

2. An 8-month-old girl weighing 6.0 kg is brought to a HC III with profuse watery diarrhoea. She is lethargic, cannot drink, has very sunken eyes and a skin pinch of 3 seconds. MUAC 12.8 cm, no oedema. An IV line is in place and Ringer's lactate is available. How should the fluid be given?

- A 180 mL over 30 minutes, then 420 mL over 2.5 hours
- B 120 mL (20 mL/kg) bolus, then maintenance fluids
- C 180 mL over 1 hour, then 420 mL over 5 hours
- D 90 mL over 1 hour, then reassess and give ReSoMal
- E 600 mL of 5% dextrose over 6 hours

3. A 2-year-old (10 kg) with diarrhoea for 5 days has passed stools with visible blood and mucus since yesterday. Temperature 38.6 °C. He is alert, drinks normally, eyes not sunken, skin pinch normal. MUAC 14 cm. Which treatment is MOST appropriate?

- A Metronidazole 100 mg 8-hourly for 7 days
- B ORS at home and zinc only; antibiotics are not recommended
- C Admit for IV ceftriaxone 80 mg/kg daily
- D Ciprofloxacin 150 mg 12-hourly for 3 days, zinc, ORS and feeding, review in 3 days
- E Loperamide to reduce the stool frequency and ORS

4. A 15-month-old girl weighing 5.8 kg (MUAC 10.8 cm, no oedema) has had watery diarrhoea for 3 days. She is irritable, eyes sunken, drinks eagerly. Hands warm, CRT 2 s, radial pulse good. Which plan is MOST appropriate?

Advanced

- A Plan B with standard ORS 435 mL over 4 hours
- B Plan C with Ringer's lactate 100 mL/kg
- C Ringer's lactate with 5% glucose 15 mL/kg over 1 hour
- D Plan A and outpatient RUTF
- E ReSoMal 5 mL/kg every 30 minutes for 2 hours, then 5–10 mL/kg/h, with F-75, as an inpatient

5. A 3-year-old (13 kg) on Plan C has received 390 mL of Ringer's lactate over 30 minutes. He is still drowsy, hands cold, radial pulse very weak. MUAC 14.5 cm. What is the BEST next step?

- A Repeat 30 mL/kg (390 mL) rapidly, check glucose, and reassess
- B Continue with 70 mL/kg over 2.5 hours as planned
- C Stop IV fluids and start NG ORS 20 mL/kg/h
- D Give furosemide 1 mg/kg because fluids are not working
- E Start dopamine immediately before any more fluid

6. During a cholera outbreak in Kasese, a 6-year-old boy (18 kg) with profuse rice-water stools has some dehydration and is drinking eagerly. Besides ORS 75 mL/kg over 4 hours, which additional treatment is recommended by the Uganda MoH 2025 cholera guidelines?

- A Metronidazole 7.5 mg/kg 8-hourly for 5 days
- B Chlorpromazine to stop the vomiting
- C Doxycycline about 50 mg (2–4 mg/kg) as a single oral dose
- D Loperamide to reduce stool output
- E Zinc 20 mg daily for 10 days and no antibiotic

7. A 9-month-old (7 kg) with acute watery diarrhoea and some dehydration vomits twice in the first 20 minutes of Plan B. She is still alert and thirsty. What is the MOST appropriate action?

- A Switch to Plan C with IV Ringer's lactate
- B Wait 10 minutes, then continue ORS more slowly with a spoon (1 teaspoon every 1–2 minutes)
- C Give metoclopramide IM and then restart ORS
- D Stop ORS and give breast milk only for 4 hours
- E Refer to the regional referral hospital

8. A 7-month-old boy has 'diarrhoea with blood' for 12 hours. His mother says he has screamed and drawn up his legs every 15–20 minutes and is pale between episodes. He vomited green fluid once. Temp 37.2 °C. What is the MOST likely diagnosis?

- A Shigella dysentery
- B Amoebic dysentery
- C Rotavirus gastroenteritis
- D Haemolytic uraemic syndrome
- E Intussusception

9. Which statement about low-osmolarity ORS compared with the old WHO ORS (311 mOsm/L) is CORRECT?

- A It has a higher sodium concentration (90 mmol/L)
- B It contains no potassium
- C It reduces stool output, vomiting and unscheduled IV therapy
- D It should not be used for cholera
- E It is the preferred fluid for children with SAM

10. An 18-month-old girl (8 kg) has had loose stools 4–5 times a day for 3 weeks. She is alert, drinks normally, no blood. WLZ −1.8, MUAC 12.8 cm, no oedema. Her mother's HIV status is unknown. Which is the MOST important next step besides feeding advice?

- A Offer HIV testing (mother and child), give vitamin A and 14 days of multivitamins with zinc, review in 5 days
- B Give metronidazole for presumed giardiasis
- C Start ciprofloxacin for 3 days
- D Admit for IV fluids
- E Stop all milk and give rice water only

11. A 2-year-old (11 kg) with SAM (MUAC 10.5 cm, oedema +) and profuse diarrhoea is lethargic with cold hands, CRT 4 s and a weak, fast pulse. RBS 4.2 mmol/L. What is the MOST appropriate immediate fluid? Advanced

- A Ringer's lactate 330 mL over 30 minutes
- B Ringer's lactate with 5% glucose 165 mL over 1 hour, then reassess
- C ReSoMal 55 mL every 30 minutes by mouth
- D Normal saline 220 mL as fast as possible, repeated twice
- E Whole blood 110 mL immediately

12. A 10-month-old (8 kg) fed on 'thick' formula (extra scoops) has diarrhoea for 3 days. He is extremely irritable with a high-pitched cry, very thirsty, and his skin feels doughy; eyes only slightly sunken. Na 162 mmol/L. Which statement about his management is MOST accurate? Advanced

A Give 5% dextrose IV to lower the sodium quickly
B Avoid ORS because it contains sodium
C His dehydration is mild because the eyes are only slightly sunken
D Rehydrate slowly so that sodium falls no faster than about 0.5 mmol/L per hour, preferring ORS if he can drink
E Restrict fluids to two-thirds maintenance

13. At a HC II with no IV equipment, a 13-month-old (9 kg, MUAC 13.4 cm) has severe dehydration. The nearest HC IV with IV fluids is 2 hours away by boda-boda. A nasogastric tube is available. What should the health worker do?

A Refer without starting any fluid, because severe dehydration needs IV therapy
B Give Plan B (75 mL/kg over 4 hours) and send home
C Start NG ORS 180 mL/h (20 mL/kg/h) for 6 hours, reassess every 1–2 hours, and refer if not improving after 3 hours
D Give ReSoMal 45 mL every 30 minutes
E Give oral zinc and ORS packets and review tomorrow

14. A 4-year-old girl (15 kg) has 3 days of watery diarrhoea without blood and has no dehydration. Her mother asks for 'a strong medicine to stop the diarrhoea'. According to the WHO 2024 guideline, which of the following is recommended?

A Amoxicillin for 5 days
B A probiotic sachet daily for 5 days
C Loperamide syrup
D Metronidazole for 5 days
E ORS, zinc and continued feeding

15. A 3-year-old treated with ciprofloxacin for Shigella dysentery 6 days ago returns pale and puffy, passing very little urine. Hb 6.2 g/dL, platelets 45 × 10^9/L, creatinine 310 µmol/L. Which finding on a blood film would MOST support the likely diagnosis? Advanced

A Target cells
B Fragmented red cells (schistocytes)
C Ring forms of P. falciparum
D Spherocytes with a positive direct Coombs test
E Hypersegmented neutrophils

Answers & explanations

> **1. A** · ORS about 700 mL over 4 hours in the clinic, zinc 20 mg daily, continue feeding, then reassess
>
> A is correct: two or more 'some dehydration' signs (restless, sunken eyes, thirsty, slow pinch) = Plan B: ORS 75 mL/kg (75 × 9.5 ≈ 710 mL) over 4 hours under supervision, zinc and feeding. B is wrong: that is Plan C, for severe dehydration (lethargy, unable to drink, very slow pinch). C is wrong: Plan A is for no dehydration; sending a thirsty, dehydrated child home under-treats him. D is wrong: ReSoMal is for children with SAM; his MUAC and feet are normal. E is wrong: there is no blood in the stool, so no antibiotic is indicated (WHO 2024 suggests against antibiotics for watery diarrhoea).

2. C · 180 mL over 1 hour, then 420 mL over 5 hours

C is correct: severe dehydration in an infant under 12 months = Plan C, Ringer's lactate 100 mL/kg (600 mL): 30 mL/kg (180 mL) in 1 hour then 70 mL/kg (420 mL) in 5 hours. A is wrong: the 30-minute/2.5-hour schedule is for children aged 12 months or older. B is wrong: a single bolus then maintenance under-replaces a >10% deficit; Plan C replaces the whole deficit. D is wrong: 15 mL/kg and ReSoMal is the SAM protocol; she is not malnourished. E is wrong: 5% dextrose contains no sodium and does not rehydrate; it can cause hyponatraemia.

3. D · Ciprofloxacin 150 mg 12-hourly for 3 days, zinc, ORS and feeding, review in 3 days

D is correct: dysentery without dehydration in a well child = ciprofloxacin 15 mg/kg (150 mg) 12-hourly for 3 days plus Plan A, zinc and follow-up in 3 days (UCG 2023, WHO 2024). A is wrong: metronidazole is for amoebiasis confirmed by trophozoites with red cells; most dysentery in under-5s is Shigella. B is wrong: WHO strongly recommends antibiotics for bloody diarrhoea. C is wrong: ceftriaxone is for severely ill children or treatment failure; this child is alert and drinking. E is wrong: antimotility drugs are contraindicated (toxic megacolon, prolonged infection).

4. E · ReSoMal 5 mL/kg every 30 minutes for 2 hours, then 5–10 mL/kg/h, with F-75, as an inpatient

E is correct: MUAC < 11.5 cm = SAM, and diarrhoea with dehydration is a complication requiring inpatient care; she is not shocked, so rehydrate orally/NG with ReSoMal slowly (5 mL/kg every 30 min for 2 h, then 5–10 mL/kg/h up to 10 h) alternating with F-75. A is wrong: standard Plan B volumes and high-sodium ORS risk heart failure in SAM (low-osmolarity ORS is only the fallback if ReSoMal is unavailable, and still given slowly). B is wrong: IV Plan C can kill a child with SAM; IV fluids only for shock. C is wrong: 15 mL/kg IV is for SAM with shock; she has warm hands and good pulses. D is wrong: a dehydrated child with SAM has a medical complication and needs inpatient stabilisation, not Plan A at home.

5. A · Repeat 30 mL/kg (390 mL) rapidly, check glucose, and reassess

A is correct: WHO Plan C says repeat the first 30 mL/kg once if the radial pulse is still very weak or undetectable; also check glucose (lethargy) and look for severe anaemia or sepsis. B is wrong: moving to the slow phase while the child is still shocked under-resuscitates him. C is wrong: NG ORS is the fallback when IV is unavailable, not for a shocked child with a working line. D is wrong: there is no fluid overload; furosemide worsens hypovolaemia. E is wrong: inotropes are considered only after adequate volume (UCG: after repeated boluses suggests septic shock), not before the repeat bolus.

6. C · Doxycycline about 50 mg (2–4 mg/kg) as a single oral dose

C is correct: the 2025 guideline gives oral antibiotics to all cholera patients; children under 12 receive doxycycline 2–4 mg/kg single dose (azithromycin 20 mg/kg single dose is the alternative). A is wrong: metronidazole has no role in cholera and is explicitly listed as a drug not to give. B is wrong: chlorpromazine (Largactil) is listed as not to be used; it causes sedation and masks signs. D is wrong: antidiarrhoeals are contraindicated. E is wrong: zinc is recommended for children under 5 (he is 6), and he still needs the antibiotic, which shortens diarrhoea and shedding.

7. B · Wait 10 minutes, then continue ORS more slowly with a spoon (1 teaspoon every 1–2 minutes)

B is correct: vomiting is common early in rehydration; the WHO/UCG advice is to wait 10 minutes and continue slowly with small, frequent sips. Most children then keep ORS down; if vomiting persists, use an NG tube at the same rate. A is wrong: she has some, not severe, dehydration; vomiting alone does not justify IV therapy. C is wrong: antiemetics are not recommended in young children with diarrhoea. D is wrong: breastfeeding should continue, but breast milk alone will not correct the deficit. E is wrong: Plan B is a HC II-level treatment and can be completed here.

8. E · Intussusception

E is correct: episodic colicky pain with pallor, red-currant-jelly stool and bilious vomiting in a 3–36-month-old is classic intussusception; request ultrasound (target sign) and a surgical review. A is wrong: Shigella usually causes fever, frequent small bloody mucoid stools and tenesmus, not episodic screaming and bilious vomiting. B is wrong: amoebic dysentery is rare under 5 years. C is wrong: rotavirus causes watery, non-bloody diarrhoea. D is wrong: HUS follows bloody diarrhoea by days and presents with pallor, oliguria and petechiae, not colic.

9. C · It reduces stool output, vomiting and unscheduled IV therapy

C is correct: low-osmolarity ORS (245 mOsm/L; Na 75, glucose 75) reduces stool volume, vomiting and the need for unscheduled IV fluids, and WHO 2024 strongly recommends it. A is wrong: sodium was reduced from 90 to 75 mmol/L. B is wrong: it contains 20 mmol/L potassium. D is wrong: it is used in cholera (and Uganda's 2025 cholera guideline even prefers standard ORS over ReSoMal in SAM children with cholera). E is wrong: ReSoMal (Na 45, K 40) is preferred in SAM; low-osmolarity ORS is the fallback.

10. A · Offer HIV testing (mother and child), give vitamin A and 14 days of multivitamins with zinc, review in 5 days

A is correct: persistent diarrhoea without dehydration is managed as an outpatient (UCG IMCI): feeding advice, vitamin A, multivitamins and minerals including zinc for 14 days, follow-up in 5 days, and in Uganda an HIV test is essential because HIV is a common cause. B is wrong: treat giardia only when seen on stool microscopy. C is wrong: there is no blood; antibiotics are not indicated. D is wrong: she is not dehydrated. E is wrong: she needs more energy and protein, not less; reduce animal milk to ≤ 50 mL/kg/day or use yoghurt, and continue nutritious foods.

11. B · Ringer's lactate with 5% glucose 165 mL over 1 hour, then reassess

B is correct: shock in SAM = 15 mL/kg (165 mL) over 1 hour of a glucose-containing fluid (RL with 5% glucose, half-strength Darrow's with 5% glucose or 0.45% saline with 5% glucose); repeat once if improving, then oral/NG ReSoMal; if not improving, blood 10 mL/kg over 3 h (UCG 2023, WHO ETAT 2016 10–15 mL/kg). A is wrong: the standard Plan C rate risks fatal heart failure in SAM. C is wrong: oral ReSoMal is for SAM without shock. D is wrong: repeated 20 mL/kg boluses are for well-nourished children. E is wrong: blood is given if there is no improvement after the first hour (or for very severe anaemia), and in SAM at 10 mL/kg over 3 hours, not as an immediate bolus.

12. D · Rehydrate slowly so that sodium falls no faster than about 0.5 mmol/L per hour, preferring ORS if he can drink

D is correct: hypernatraemic dehydration must be corrected slowly (≤ 0.5 mmol/L/h, about 10–12 mmol/L in 24 h, over 48 h or more) to avoid cerebral oedema; NICE and WHO advise ORS even in hypernatraemia unless IV is needed. Also correct the feeding error. A is wrong: hypotonic dextrose lowers sodium too fast and causes convulsions. B is wrong: ORS is safe and preferred. C is wrong: hypernatraemia masks clinical signs, so the deficit is underestimated. E is wrong: he is dehydrated; restriction is for SIADH-type hyponatraemia, not this.

13. C · Start NG ORS 180 mL/h (20 mL/kg/h) for 6 hours, reassess every 1–2 hours, and refer if not improving after 3 hours

C is correct: when IV therapy is not available within 30 minutes, start ORS by NG tube (or mouth) at 20 mL/kg/h for 6 hours (120 mL/kg), reassess every 1–2 hours and refer urgently if not improving by 3 hours. A is wrong: a 2-hour journey without fluid can be fatal; start rehydration now (and if referring, the mother gives ORS on the way). B is wrong: Plan B under-treats severe dehydration. D is wrong: ReSoMal is for SAM, which he does not have. E is wrong: it ignores a life-threatening emergency.

14. E · ORS, zinc and continued feeding

E is correct: WHO 2024 recommends low-osmolarity ORS and zinc and suggests against antibiotics and probiotics for acute watery diarrhoea in children up to 10 years. A is wrong: antibiotics do not help acute watery diarrhoea and drive resistance. B is wrong: WHO suggests against probiotics for acute watery diarrhoea. C is wrong: antimotility drugs are not recommended in children. D is wrong: metronidazole is only for proven amoebiasis or giardiasis.

15. B · Fragmented red cells (schistocytes)

B is correct: anaemia, thrombocytopenia and AKI after bloody diarrhoea = haemolytic uraemic syndrome, a microangiopathic haemolytic anaemia with schistocytes (S. dysenteriae type 1 or EHEC). Refer for dialysis and careful fluid management. A is wrong: target cells suggest haemoglobinopathy or liver disease. C is wrong: malaria can cause anaemia and AKI, but the sequence after dysentery with thrombocytopenia and AKI points to HUS (still do an RDT). D is wrong: Coombs-positive spherocytosis suggests autoimmune haemolysis, and HUS is Coombs-negative. E is wrong: hypersegmented neutrophils indicate B12/folate deficiency.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Severe dehydration in a toddler at a HC IV

You are the intern at a HC IV in Wakiso at 21:00. Brian, a 20-month-old boy, is carried in by his mother. He has had watery diarrhoea about 12 times a day and vomiting for 2 days. Today he became 'weak and sleepy' and has not passed urine since the morning.

1. What do you do in the first 5 minutes?

> **ETAT triage**: he is lethargic = emergency. Check **ABC**: airway open, RR 48 deep and sighing (acidosis), SpO_2 97%. **Circulation**: cold hands, **CRT 4 s**, weak fast radial pulse (HR 170) = **shock**. **Weigh him: 10 kg.** Check **MUAC (13.6 cm) and feet (no oedema)**: not SAM. Insert an IV (or IO) line, take blood for RBS (4.0 mmol/L), Hb (11 g/dL) and malaria RDT, and **start Ringer's lactate at once**. Call the senior clinician.

2. Classify his hydration and write the fluid prescription.

> Lethargic, very sunken eyes, unable to drink, skin pinch 3 s = **severe dehydration with shock**, no SAM. He is ≥ 12 months: **Plan C, Ringer's lactate 100 mL/kg = 1000 mL**:
>
> - **300 mL (30 mL/kg) in 30 minutes** (run free, squeeze the bag)
> - reassess; if radial pulse still very weak, **repeat 300 mL once**
> - then **700 mL (70 mL/kg) over 2.5 hours** (280 mL/h)
> - **ORS 50 mL/h (5 mL/kg/h)** once he can drink
> - Observations every 15 min initially.

3. After the first 300 mL his pulse is stronger, CRT 2 s, and he is waking. What do you monitor during the next 2.5 hours, and what would make you stop?

> Every 15–30 min: **conscious level, pulse rate and volume, CRT, RR, ability to drink, urine output, stool and vomit count**, and recheck glucose if drowsy. Stop or slow the drip and call the senior if signs of **overload**: crackles, gallop, enlarging liver, puffy eyelids, rising RR with falling SpO_2. Keep a fluid chart with times and volumes.

4. At 3 hours he is alert, drinks eagerly, eyes still a little sunken, skin pinch slow. What now?

> Reclassify: **some dehydration → Plan B**: ORS **75 mL/kg = 750 mL over 4 hours**, then reassess. Start **zinc 20 mg once daily for 10 days**. Restart **breastfeeding and food**. Malaria RDT was negative, so no antimalarial.

5. Next morning he has no dehydration. What do you teach the mother before discharge?

> **Plan A (4 rules)**: ORS 50–100 mL after each loose stool (he is under 2), how to mix **1 packet in 1 L**, discard after 24 h; **complete zinc for 10 days**; continue breastfeeding and feeding, extra meal for 2 weeks; **return immediately** if he cannot drink, becomes sicker, has fever or blood in stool. Check his card: catch up any missed vaccines (rotavirus is only given early; measles–rubella if due). Hygiene: hand-washing with soap, safe water, latrine use.

6. How do you present him on the ward round?

> 'Brian, a 20-month-old boy, 10 kg, MUAC 13.6 cm, no oedema, fully immunised, HIV-unexposed, admitted with acute watery diarrhoea and severe dehydration with shock. He received Plan C (1000 mL RL; the first 30 mL/kg once only), then Plan B, and is now fully rehydrated, passing urine, feeding and on zinc. Malaria RDT negative. Plan: Plan A, discharge with ORS and zinc, review in 5 days if not better.'

Take-away: Weigh and check for SAM first, then give Plan C by the clock (30 then 70 mL/kg), reassess often, step down through Plan B to Plan A, and never forget zinc and feeding.

Case 2: Diarrhoea in a child with severe acute malnutrition

Amina, a 13-month-old girl, is referred from a HC III in Mayuge to Mengo Hospital with 4 days of watery diarrhoea. The referral note says 'severe dehydration, please give IV fluids'. She weighs 5.9 kg, length 72 cm.

1. What must you check before deciding on fluids?

Nutritional status: MUAC = **10.6 cm**, WLZ **below −3**, visible severe wasting; **no oedema**. She has **SAM with a complication** (diarrhoea). Then **circulation**: hands warm, CRT 2 s, radial pulse fair, so **not in shock**. Also **RBS** (2.6 mmol/L = hypoglycaemia), **temperature** (axillary 35.4 °C), Hb, malaria RDT, HIV test.

2. How do you treat the hypoglycaemia and hypothermia?

Give **50 mL of 10% glucose or sugar water orally/NG** (she is conscious) and start **F-75** feeds; if she were unconscious, **10% glucose 5 mL/kg IV**. Recheck glucose in **30 minutes**. Warm her: **skin-to-skin (kangaroo) with mother**, cover head, close windows; no hot water bottles. Start **IV/IM antibiotics** (ampicillin + gentamicin), because hypoglycaemia and hypothermia signal infection.

3. Write her rehydration plan.

No IV fluids (not shocked). **ReSoMal** orally or by NG:

- **5 mL/kg (30 mL) every 30 minutes for 2 hours**
- then **5–10 mL/kg/h (30–60 mL/h) for up to 10 hours**, alternating hours with **F-75** in the same volume
- if ReSoMal is unavailable, **low-osmolarity ORS** (WHO 2023) or half-strength ORS with added glucose and potassium (UCG)
- **ReSoMal 50–100 mL after each watery stool** afterwards
- **No zinc tablets** (F-75 contains zinc).

4. What do you monitor and when do you stop?

Every **30 minutes for 2 hours**, then **hourly**: pulse, RR, **liver edge**, urine frequency, stools and vomits, **weight**. **Stop ReSoMal** if pulse rises by **25/min** and RR by **5/min**, or the liver enlarges, crackles or oedema appear (overload); reassess after 1 hour. Stop when she is **no longer thirsty, passing urine**, and the **weight deficit is corrected**: do not keep giving fluid 'to prevent' dehydration.

5. Six hours later her pulse rises from 140 to 168, RR from 40 to 48, and the liver edge is 2 cm lower. What is happening and what do you do?

Fluid overload / early heart failure, the feared complication of rehydrating a child with SAM. **Stop all fluids**, sit her up, give **furosemide 1 mg/kg IV**, oxygen if hypoxic, and reassess in 1 hour; then continue F-75 at **100 mL/kg/day** and increase slowly. Call the senior.

6. Why was the referral note's request for IV fluids wrong, and what will you write back to the HC III?

In SAM, signs of dehydration are unreliable (wasting itself causes sunken eyes and slow skin pinch), the heart is weak and sodium is retained, so **IV fluids are only for shock** (15 mL/kg over 1 h). Thank them for referring; feed back: 'Always measure MUAC and check for oedema in a child with diarrhoea; if SAM and not shocked, give ReSoMal 5 mL/kg every 30 min for 2 h while arranging referral, and keep the child warm and fed.'

Take-away: In SAM, dehydration is over-diagnosed and IV fluids kill: rehydrate slowly with ReSoMal, treat hypoglycaemia, hypothermia and infection, and watch the pulse, breathing and liver for overload.

Case 3: Bloody diarrhoea in a febrile toddler

Joel, a 2½-year-old boy (12 kg), is brought to the paediatric outpatient clinic at a regional referral hospital with 2 days of frequent small stools containing blood and mucus, fever and crying when he passes stool. His sister had similar symptoms last week.

1. What history and examination will you focus on?

> History: number of stools, **blood and mucus**, **tenesmus**, fever, **convulsions**, urine output, vomiting, drinking, previous treatment (any antibiotics or herbs), **HIV status**, immunisation, feeding, water source, contacts. Exam: **dehydration signs**, temperature (39 °C), **MUAC (14 cm)**, oedema, abdomen (tenderness, mass), **rectal prolapse**, neck stiffness, pallor, petechiae. His hydration: alert, drinking normally, eyes normal = **no dehydration**.

2. What is your classification and what are your differentials?

> **Dysentery** (UCG IMCI) without dehydration. Likely **Shigella** (fever, tenesmus, household contact). Differentials: **Campylobacter**, **non-typhoidal Salmonella**, EHEC, **amoebic dysentery** (rarer at this age), **intussusception** (episodic pain, mass, red-currant jelly), and malaria with diarrhoea.

3. What tests, if any, will you do?

> **Malaria RDT** (fever). Stool **microscopy** (pus cells; look for amoebic trophozoites with ingested red cells) and ideally **stool culture** given the outbreak pattern. **FBC** if he looks unwell. RBS if drowsy or not eating. Treatment does **not** wait for results.

4. Prescribe his treatment.

> - **Ciprofloxacin 15 mg/kg = 180 mg 12-hourly for 3 days** (in practice the UCG age band gives ½ of a 500 mg tablet for 6 months–5 years)
> - **Zinc 20 mg once daily for 10 days**
> - **Plan A**: ORS 100–200 mL after each stool (he is over 2)
> - **Paracetamol 15 mg/kg (180 mg) 6-hourly** as needed
> - Continue feeding, **no antimotility drugs**
> - **Follow up in 3 days**.

5. He returns after 2 days: still bloody stools, fever 39.5 °C, now drinking poorly, eyes sunken, skin pinch slow. What now?

> **Treatment failure with some dehydration**: admit. **Plan B** (ORS 75 mL/kg = 900 mL over 4 h, NG if not drinking). Switch to **ceftriaxone 50–80 mg/kg (600–960 mg) IV once daily for 3 days**. Send **stool culture and sensitivity**; look for **amoebae** (metronidazole 10 mg/kg 8-hourly if trophozoites with RBCs are seen). Check **glucose, Hb, platelets, creatinine** and urine output; watch for **convulsions and HUS**.

6. What public-health actions follow?

> Bacillary dysentery is **notifiable**: inform the district surveillance focal person, especially with a household cluster. Counsel the family on **hand-washing with soap**, safe water, latrine use, safe food handling, and caring for the sister. Ensure everyone at home knows the **danger signs**.

Take-away: Blood in the stool means an antibiotic (ciprofloxacin 15 mg/kg 12-hourly for 3 days), close follow-up at 2–3 days, a switch to ceftriaxone if not improving, and vigilance for dehydration, HUS and intussusception.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.